MTH 132 Midterm 2013

Name______________________________ Date__________________________

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

Use the given conditions to write an equation for the line in slope-intercept form.

1) Slope = –3, passing through (6, 6) 1) ________

A) y – 6 = –3x – 6 B) y – 6 = x – 6 C) y = –3x + 24 D) y = –3x – 24

2) Slope = 4, passing through (7, 3) 2) ________

A) y = 4x – 25 B) y – 3 = 4x – 7 C) y = 4x + 25 D) y – 3 = x – 7

Find the equation of the least squares line.

3) The paired data below consist of the test scores of 6 randomly selected students and the number of hours they studied for the test. 3) ________

Hours (x)	5	10	4	6	10	9
Score (y)	64	86	69	86	59	87

A) y = 33.7 + 2.14x B) y = 67.3 + 1.07x

C) y = 33.7 – 2.14x D) y = –67.3 + 1.07x

Write a cost function for the problem. Assume that the relationship is linear.

4) A moving firm charges a flat fee of $40 plus $35 per hour. Let C(x) be the cost in dollars of using the moving firm for x hours. 4) ________

A) C(x) = 40x – 35 B) C(x) = 35x – 40 C) C(x) = 35x + 40 D) C(x) = 40x + 35

Solve the system.

5) x + y = –2 5) ________
x – y = 18

A) {(8, –10)} B) {(8, 10)} C) {(2, –10)} D) {(2, 8)}

Solve the system of two equations in two unknowns.

6) x + 6y = –5 6) ________
–4x + 7y = 20

A) (–5, 0) B) (–4, –5) C) (5, –1) D) No solution

Find the slope of the line passing through the given pair of points.

7) (3, 4) and (3, −8) 7) ________

A) $-\frac{2}{3}$ B) Undefined C) 0 D) 2

8) (−4, 1) and (−9, 1) 8) ________

A) Not defined B) 0 C) $-\frac{2}{13}$ D) $-\frac{2}{5}$

9) (5, −5) and (−7, −3) 9) ________

A) $-\frac{1}{6}$ B) 4 C) − 6 D) $\frac{1}{6}$

10) (6, 5) and (−4, 5) 10) ________

A) 5 B) Undefined C) − 1 D) 0

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA, whenever these products exist.

11) A is 1 × 3, and B is 3 × 1. 11) ________

A) AB is 1 × 1; BA is 3 × 3. B) AB is 3 × 3; BA is 1 × 1.
C) AB does not exist; BA is 3 × 3. D) AB is 1 × 1; BA does not exist.

Solve the problem.

12) 12) ________

Let $A = \begin{bmatrix} -1 & 0 \\ 6 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 6 \\ 3 & 1 \end{bmatrix}$. Find A − B.

A) $\begin{bmatrix} 0 & 6 \\ -3 & -2 \end{bmatrix}$ B) $\begin{bmatrix} 0 & -6 \\ 3 & 2 \end{bmatrix}$ C) $\begin{bmatrix} -1 \end{bmatrix}$ D) $\begin{bmatrix} -2 & 6 \\ 9 & 4 \end{bmatrix}$

Find the matrix product, if possible.

13) $\begin{bmatrix} 1 & 3 & -2 \\ 2 & 0 & 3 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ -2 & 1 \\ 0 & 3 \end{bmatrix}$ 13) ________

A) $\begin{bmatrix} 3 & -6 & 0 \\ 0 & 0 & 9 \end{bmatrix}$ B) $\begin{bmatrix} -3 & -3 \\ 6 & 9 \end{bmatrix}$ C) $\begin{bmatrix} -3 & -3 \\ 9 & 6 \end{bmatrix}$ D) Does not exist

Solve the problem.

14) Let $A = \begin{bmatrix} -3 & 2 \\ 0 & 2 \end{bmatrix}$. Find 4A.

14) ________

A) $\begin{bmatrix} -12 & 2 \\ 0 & 2 \end{bmatrix}$

B) $\begin{bmatrix} 1 & 6 \\ 4 & 6 \end{bmatrix}$

C) $\begin{bmatrix} -12 & 8 \\ 0 & 8 \end{bmatrix}$

D) $\begin{bmatrix} -12 & 8 \\ 0 & 2 \end{bmatrix}$

15) Let $A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 4 \\ -1 & 6 \end{bmatrix}$. Find 2A + B.

15) ________

A) $\begin{bmatrix} 4 & 10 \\ 1 & 10 \end{bmatrix}$

B) $\begin{bmatrix} 4 & 14 \\ 2 & 20 \end{bmatrix}$

C) $\begin{bmatrix} 4 & 7 \\ 3 & 10 \end{bmatrix}$

D) $\begin{bmatrix} 4 & 10 \\ 3 & 14 \end{bmatrix}$

16) Let $A = \begin{bmatrix} -5 & 1 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} 6 & 2 \\ 3 & 2 \end{bmatrix}$. Find A + B.

16) ________

A) $\begin{bmatrix} 1 & 3 \\ 5 & 7 \end{bmatrix}$

B) $\begin{bmatrix} 16 \end{bmatrix}$

C) $\begin{bmatrix} 3 & 4 \\ 3 & 7 \end{bmatrix}$

D) $\begin{bmatrix} 1 & -3 \\ -2 & -3 \end{bmatrix}$

Find the matrix product, if possible.

17) $\begin{bmatrix} 3 & -2 & 1 \\ 0 & 4 & -1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ -2 & 3 \end{bmatrix}$

17) ________

A) $\begin{bmatrix} 9 & -6 & 3 \\ -6 & 16 & -5 \end{bmatrix}$

B) $\begin{bmatrix} 9 & -6 \\ -6 & 16 \\ 3 & -5 \end{bmatrix}$

C) $\begin{bmatrix} 9 & 0 \\ 0 & 12 \end{bmatrix}$

D) Does not exist

Find the inverse, if it exists, for the matrix.

18) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -2 & 3 & 1 \end{bmatrix}$

18) ________

A) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -5 & 4 & 1 \end{bmatrix}$

B) No Inverse

C) $\begin{bmatrix} 2 & -1 & 0 \\ 3 & -2 & 0 \\ -5 & 4 & -1 \end{bmatrix}$

D) $\begin{bmatrix} 1 & -1 & 0 \\ 3 & -2 & 1 \\ -5 & 4 & -1 \end{bmatrix}$

Solve the problem.

19) A digital pen manufacturer has decided to come out with a new and affordable digital pen . The fixed cost for the production of this new digital pen line is $16,600 and the variable costs are $68 per digital pen . The company expects to sell the digital pens for $159. Formulate a function C(x) for the total cost of producing x new digital pens and a function R(x) for the total revenue generated from the sales of x digital pens. 19) ________

A) $C(x) = 16600 + 159x$; $R(x) = 68x$
B) $C(x) = 16600 + 68x$; $R(x) = 159x$
C) $C(x) = 16,668$; $R(x) = 159$
D) $C(x) = 68x$; $R(x) = 159x$

20) There were 570 people at a school fundraising concert. The admission price was $2 for adults and $1 for children. The admission receipts were $830. How many adults and how many children attended? 20) ________

A) 207 adults, 363 children
B) 155 adults, 415 children
C) 310 adults, 260 children
D) 260 adults, 310 children

21) Sweet Honey Bakery serves desserts. One serving of ice cream and two servings of blueberry pie provides 820 calories. Three servings of ice cream and two servings of blueberry pie provides 1300 calories. Find the caloric content of each item. 21) ________

A) Serving of frozen yogurt: 240 calories
Serving of apple cobbler: 290 calories
B) Serving of frozen yogurt: 290 calories
Serving of apple cobbler: 240 calories
C) Serving of frozen yogurt: 252 calories
Serving of apple cobbler: 284 calories
D) Serving of frozen yogurt: 234 calories
Serving of apple cobbler: 293 calories

22) After two years on the job, a therapist's salary was $60,000. After seven years on the job, her salary was $72,500. Let y represent her salary after x years on the job. Assuming that the change in her salary over time can be approximated by a straight line, give an equation for this line in the form $y = mx + b$. 22) ________

A) $y = 12,500x + 60,000$
B) $y = 2500x + 60,000$
C) $y = 12,500x + 35,000$
D) $y = 2500x + 55,000$

23) FedUps delivers packages which cost $2.00 per package to deliver. The fixed cost to run the delivery truck is $234 per day. If the company charges $5.00 per package, how many packages must be delivered daily to make a profit of $42? 23) ________

A) 33 packages
B) 117 packages
C) 78 packages
D) 92 packages

24) A shoe company will make a new design of walking shoe. The fixed cost for the production will be $24,000. The variable cost will be $31 per pair of shoes. The shoes will sell for $106 for each pair. How many pairs of shoes will have to be sold for the company to break even on this new line of shoes? 24) _______

A) 321 pairs B) 775 pairs C) 75 pairs D) 227 pairs

25) Let the supply and demand functions for gourmet hot chocolate packs be given by 25) _______

$$p = S(q) = \frac{5}{4}q \quad \text{and} \quad p = D(q) = 70 - \frac{3}{4}q,$$

where p is the price in dollars and q is the number of batches.. Find the equilibrium quantity and the equilibrium price.

A) Equilibrium quantity: 35
Equilibrium price: $43.75

B) Equilibrium quantity: 35.00
Equilibrium price: $28

C) Equilibrium quantity: 28
Equilibrium price: $35.00

D) Equilibrium quantity: 43.75
Equilibrium price: $35

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

26) In one or more paragraphs, explain what it means to "break even" in business. 26) _______________

Name_______________________________ Date_______________________________

Find the matrix product, if possible.

1) $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 5 & 2 & -1 \\ 2 & -1 & 2 \end{bmatrix}$

A) $\begin{bmatrix} 6 & -3 & 6 \\ 5 & 2 & -1 \end{bmatrix}$ B) $\begin{bmatrix} 5 & 0 & 0 \\ 0 & -3 & 6 \end{bmatrix}$

C) $\begin{bmatrix} 5 & 2 & -1 \\ 6 & -3 & 6 \end{bmatrix}$ D) Does not exist

Use any method to solve the system of two equations in two unknowns.

2) $x + 5y = 20$
$7x + 6y = 24$

A) (0, 4) B) (1, 3)
C) (−4, 0) D) No solution

Find the number of subsets of the set.

3) {math, English, history, science, art}

A) 24 B) 16 C) 32 D) 28

Use the given conditions to write an equation for the line in slope-intercept form.

4) Slope = 2, passing through (7, 2)

A) $y - 2 = 2x - 7$ B) $y = 2x + 12$
C) $y - 2 = x - 7$ D) $y = 2x - 12$

Solve the problem.

5) Let $C = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ and $D = \begin{bmatrix} -1 \\ 3 \\ -2 \end{bmatrix}$. Find $C - 3D$.

A) $\begin{bmatrix} -4 \\ 12 \\ -8 \end{bmatrix}$ B) $\begin{bmatrix} 4 \\ -12 \\ 8 \end{bmatrix}$

C) $\begin{bmatrix} -2 \\ 6 \\ -4 \end{bmatrix}$ D) $\begin{bmatrix} 4 \\ -6 \\ 4 \end{bmatrix}$

Write a negation for the statement.

6) My brother is asleep.

A) My sister is awake.
B) My sister is asleep.
C) My brother is not asleep.
D) The person who is asleep is not my brother.

Solve the problem.

7) There were 670 people at a play. The admission price was \$3 for adults and \$1 for children. The admission receipts were \$1310. How many adults and how many children attended?

A) 320 adults, 350 children
B) 350 adults, 320 children
C) 327 adults, 343 children
D) 175 adults, 495 children

Find the equation of the least squares line.

8) The paired data below consist of the costs of advertising (in thousands of dollars) and the number of products sold (in thousands).

Cost (x)	9	2	3	4	2	5	9	10
Number (y)	85	52	55	68	67	86	83	73

A) $y = 55.8 + 2.79x$
B) $y = -26.4 - 1.42x$
C) $y = 26.4 + 1.42x$
D) $y = 55.8 - 2.79x$

Find the matrix product, if possible.

9) $\begin{bmatrix} 3 & -2 & 1 \\ 0 & 4 & -1 \end{bmatrix}\begin{bmatrix} 3 & 0 \\ -2 & 3 \end{bmatrix}$

A) $\begin{bmatrix} 9 & -6 & 3 \\ -6 & 16 & -5 \end{bmatrix}$ B) $\begin{bmatrix} 9 & 0 \\ 0 & 12 \end{bmatrix}$

C) $\begin{bmatrix} 9 & -6 \\ -6 & 16 \\ 3 & -5 \end{bmatrix}$ D) Does not exist

Find the simple interest. Assume a 360-day year. Round results to the nearest cent.

10) \$16,150 at 11.38% for 76 days

A) \$16,150.00 B) \$387.82
C) \$382.72 D) \$1837.06

Find the slope of the line passing through the given pair of points.

11) (2, −5) and (9, −5)

A) 0 B) $-\dfrac{10}{7}$

C) Undefined D) $-\dfrac{10}{11}$

Write an equation for the line. Use slope-intercept form, if possible.

12) Through (4, 0) and (−8, 7)

A) $y = \frac{7}{12}x + \frac{7}{3}$ B) $y = \frac{4}{15}x + \frac{73}{15}$

C) $y = -\frac{4}{15}x + \frac{73}{15}$ D) $y = -\frac{7}{12}x + \frac{7}{3}$

13) Through (13, 5), m = 2

A) $y = -2x + 3$ B) $y = 2x + 5$

C) $y = -2x + 21$ D) $y = 2x - 21$

14) Through (−3, 0) and (−6, 4)

A) $y = \frac{3}{10}x + \frac{29}{5}$ B) $y = \frac{4}{3}x - 4$

C) $y = -\frac{3}{10}x + \frac{29}{5}$ D) $y = -\frac{4}{3}x - 4$

Find the amount that should be invested now to accumulate the following amount, if the money is compounded as indicated.

15) $5000 at 5% compounded quarterly for 3 yr

A) $5803.77 B) $4319.19

C) $4307.54 D) $692.46

Use any method to solve the system of two equations in two unknowns.

16) $9x + 7y = 88$
$-3x + 4y = -23$

A) (8, 2) B) (9, 2)

C) (9, 1) D) No solution

Find the inverse, if it exists, for the matrix.

17) $\begin{bmatrix} 0 & 4 \\ 3 & 4 \end{bmatrix}$

A) $\begin{bmatrix} -\frac{1}{3} & \frac{1}{3} \\ \frac{1}{4} & 0 \end{bmatrix}$ B) $\begin{bmatrix} -\frac{1}{3} & -\frac{1}{3} \\ -\frac{1}{4} & 0 \end{bmatrix}$

C) $\begin{bmatrix} \frac{1}{4} & 0 \\ -\frac{1}{3} & \frac{1}{3} \end{bmatrix}$ D) $\begin{bmatrix} 0 & \frac{1}{3} \\ \frac{1}{4} & -\frac{1}{3} \end{bmatrix}$

Solve the problem.

18) Let the supply and demand functions for raspberry-flavored licorice be given by

$$p = S(q) = \frac{5}{4}q \quad \text{and} \quad p = D(q) = 70 - \frac{5}{4}q,$$

where p is the price in dollars and q is the number of batches.. Find the equilibrium quantity and the equilibrium price.

A) Equilibrium quantity: 35.00
Equilibrium price: $28

B) Equilibrium quantity: 35
Equilibrium price: $43.75

C) Equilibrium quantity: 43.75
Equilibrium price: $35

D) Equilibrium quantity: 28
Equilibrium price: $35.00

19) After two years on the job, an engineer's salary was $60,000. After seven years on the job, her salary was $66,000. Let y represent her salary after x years on the job. Assuming that the change in her salary over time can be approximated by a straight line, give an equation for this line in the form y = mx + b.

A) y = 1200x + 57,600

B) y = 6000x + 60,000

C) y = 6000x + 48,000

D) y = 1200x + 60,000

20) The change in a certain engineer's salary over time can be approximated by the linear equation y = 1500x + 47,500 where y represents salary in dollars and x represents number of years on the job. According to this equation, after how many years on the job was the engineer's salary $64,000?

A) 10 years B) 12 years

C) 11 years D) 13 years

Perform the indicated operation where possible.

21) $\begin{bmatrix} 2 & 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 9 \end{bmatrix}$

A) $\begin{bmatrix} 3 \\ 13 \end{bmatrix}$ B) $\begin{bmatrix} 2 & 1 \\ 4 & 9 \end{bmatrix}$

C) $\begin{bmatrix} 3 & 13 \end{bmatrix}$ D) Not possible

22) $\begin{bmatrix} -1 & 4 \\ 0 & 4 \\ 9 & -4 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 7 & 4 \\ 4 & 2 \end{bmatrix}$

A) $\begin{bmatrix} 1 & 5 \\ 7 & 8 \\ 13 & -1 \end{bmatrix}$ B) $\begin{bmatrix} -3 & 3 \\ -7 & 0 \\ 5 & -6 \end{bmatrix}$

C) $\begin{bmatrix} 3 & -3 \\ 7 & 0 \\ -5 & 6 \end{bmatrix}$ D) Not possible

23) $\begin{bmatrix} -4 & 1 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} 6 & 2 \\ 1 & -2 \end{bmatrix}$

A) $\begin{bmatrix} 3 & 4 \\ -1 & 3 \end{bmatrix}$ B) $\begin{bmatrix} 2 & -2 \\ -3 & -6 \end{bmatrix}$

C) $\begin{bmatrix} 2 & 3 \\ 3 & 3 \end{bmatrix}$ D) Not possible

The sizes of two matrices A and B are given. Find the sizes of the product AB and the product BA, whenever these products exist.

24) A is 1×2, and B is 2×1.

A) AB is 1×1; BA is 2×2.
B) AB is 1×1; BA does not exist.
C) AB does not exist; BA is 2×2.
D) AB is 2×2; BA is 1×1.

Find the slope of the line passing through the given pair of points.

25) (4, 9) and (2, 6)

A) $-\frac{3}{2}$ B) $\frac{5}{2}$

C) $\frac{2}{3}$ D) $\frac{3}{2}$

26) (4, 2) and (4, 9)

A) $-\frac{7}{8}$ B) $\frac{11}{8}$

C) 0 D) Undefined

Write the augmented matrix for the system. Do not solve.

27) $7x + 9y = 40$
$6x + 5y = 37$

A) $\left[\begin{array}{cc|c} 40 & 9 & 7 \\ 37 & 6 & 5 \end{array}\right]$ B) $\left[\begin{array}{cc|c} 7 & 6 & 40 \\ 9 & 5 & 37 \end{array}\right]$

C) $\left[\begin{array}{cc|c} 7 & 9 & 40 \\ 6 & 5 & 37 \end{array}\right]$ D) $\left[\begin{array}{cc|c} 7 & 9 & 37 \\ 5 & 6 & 40 \end{array}\right]$

Find the effective rate corresponding to the given nominal rate. Round results to the nearest 0.01 percentage points.

28) 2% compounded monthly

A) 0.27% B) 1.02%
C) 2.01% D) 2.02%

Find the compound amount for the deposit. Round to the nearest cent.

29) $3900 at 11% compounded semiannually for 6 years

A) $7294.62 B) $6474.00
C) $7414.71 D) $5377.49

Write a cost function for the problem. Assume that the relationship is linear.

30) A moving firm charges a flat fee of $35 plus $ 30 per hour. Let C(x) be the cost in dollars of using the moving firm for x hours.

A) $C(x) = 30x - 35$ B) $C(x) = 35x + 30$
C) $C(x) = 30x + 35$ D) $C(x) = 35x - 30$

Let U = {q, r, s, t, u, v, w, x, y, z}; A = {q, s, u, w, y}; B = {q, s, y, z}; and C = {v, w, x, y, z}. List the members of the indicated set, using set braces.

31) $A' \cup B$

A) {q, s, t, u, v, w, x, y}
B) {q, r, s, t, v, x, y, z}
C) {r, s, t, u, v, w, x, z}
D) {s, u, w}

32) $A \cup (B \cap C)$

A) {q, w, y} B) {q, r, w, y, z}
C) {q, y, z} D) {q, s, u, w, y, z}

Decide whether the statement is true or false.

33) $\{4, 8, 12, 16\} \cup \{4, 12\} = \{4, 8, 12, 16\}$

Let p represent a true statement, and let q and r represent false statements. Find the truth value of the given compound statement.

34) $(\sim p \wedge \sim q) \vee \sim q$

A) True B) False

35) $(p \wedge \sim q) \wedge r$

A) False B) True

Translate the symbolic compound statement into words.

36) Let p represent the statement "Students are males" and let q represent the statement "Teachers are males."

$\sim(p \vee \sim q)$

A) It is not the case that students are males and teachers are not males.
B) Students are not males or teachers are not males.
C) It is not the case that students are males or teachers are not males.
D) Students are not males and teachers are not males.

Solve the problem.

37) A shoe company will make a new type of shoe. The fixed cost for the production will be \$24,000. The variable cost will be \$35 per pair of shoes. The shoes will sell for \$101 for each pair. What is the profit if 600 pairs are sold?

A) \$63,600 B) \$57,600
C) \$39,600 D) \$15,600

38) Regrind, Inc. regrinds used typewriter platens. The cost per platen is \$1.80. The fixed cost to run the grinding machine is \$112 per day. If the company sells the reground platens for \$ 3.80, how many must be reground daily to break even?

A) 56 platens B) 62 platens
C) 37 platens D) 20 platens

Use a Venn diagram to answer the question.

39) At East Zone University (EZU) there are 523 students taking Math or English. 163 are taking Math, 399 are taking English, and 39 are taking both . How many are taking English but not Math ?

A) 360 B) 484
C) 85 D) 124

Decide whether the following is a statement or is not a statement.

40) 10 is greater than 12.

A) Not a statement
B) Statement

41) ***Suppose you poll the students in your class and find that 11 have sisters and 14 have brothers. Why is it not correct to assume that there are only 25 students in the class? Discuss at least two possibilities of what the total number of persons could be in your class(other than 25)***